

The Fundamental Theorem of Calculus

ANSWER KEY



Part 2: evaluating definite integrals

- 1 Evaluate each definite integral:

a $\int_1^3 (2x + 1) dx \quad [x^2 + x]_1^3 = 12 - 2 = 10.$

b $\int_0^\pi \sin x dx \quad [-\cos x]_0^\pi = 1 - (-1) = 2.$

c $\int_1^4 \sqrt{x} dx \quad \left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}.$

- 2 Evaluate $\int_0^1 e^{2x} dx$, to three decimal places.
 $\left[\frac{1}{2}e^{2x}\right]_0^1 = \frac{1}{2}(e^2 - 1) \approx 3.195.$

Part 1: differentiating an integral

- 3 Use the Fundamental Theorem, Part 1, to find $\frac{d}{dx} \int_2^x (t^3 + 1) dt$.
 $= x^3 + 1$ (directly, by FTC Part 1).
- 4 Find $\frac{d}{dx} \int_1^{x^2} \sin(t) dt$ using the chain rule with FTC.
 $= \sin(x^2) \cdot 2x = 2x \sin(x^2).$
- 5 Let $g(x) = \int_{\sqrt{x}}^x \sqrt{4 + t^2} dt$. Find $g'(3)$.
 $g'(x) = \sqrt{4 + x^2}$, so $g'(3) = \sqrt{13}.$

Average value and applications

- 6 Find the average value of $f(x) = x^2$ on $[0, 3]$.
 $\frac{1}{3} \int_0^3 x^2 dx = \frac{1}{3} \cdot 9 = 3.$
- 7 A car's velocity is $v(t) = 3t^2 + 2$ m/s for $0 \leq t \leq 4$. Find the total distance traveled, and the average velocity over this interval.
Distance $= \int_0^4 (3t^2 + 2) dt = [t^3 + 2t]_0^4 = 64 + 8 = 72$ m. Average velocity $= \frac{72}{4} = 18$ m/s.

Further FTC practice

- 8 Evaluate $\int_0^2 (x^3 - 2x + 3) dx$.
 $\left[\frac{x^4}{4} - x^2 + 3x\right]_0^2 = (4 - 4 + 6) - 0 = 6.$
- 9 Find $\frac{d}{dx} \int_x^5 \cos(t^2) dt$, using the property $\int_a^b = -\int_b^a$ before applying FTC Part 1.
 $\int_x^5 \cos(t^2) dt = -\int_5^x \cos(t^2) dt$, so the derivative is $-\cos(x^2).$

- 10 Find the average value of $f(x) = \sin x$ on $[0, \pi]$, to three decimal places.

$$\frac{1}{\pi} \int_0^{\pi} \sin x \, dx = \frac{1}{\pi} [-\cos x]_0^{\pi} = \frac{2}{\pi} \approx 0.637.$$

Mean Value Theorem for integrals

- 11 Find the value(s) of c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$ (where $f(c)$ equals the average value of f).

Average value $= \frac{1}{3} \int_0^3 x^2 \, dx = 3$ (found earlier). Solve $c^2 = 3$: $c = \sqrt{3}$ (taking the positive root within $[0, 3]$).

- 12 Evaluate $\int_2^5 f(x) \, dx$ given that the average value of f on $[2, 5]$ is 7.

Average value $= \frac{1}{5-2} \int_2^5 f(x) \, dx = 7$, so $\int_2^5 f(x) \, dx = 7(3) = 21$.

Integrals of piecewise functions

- 13 Evaluate $\int_0^4 f(x) \, dx$ where $f(x) = x$ for $x \leq 2$ and $f(x) = 2x - 2$ for $x > 2$.

$$\int_0^2 x \, dx + \int_2^4 (2x - 2) \, dx = \left[\frac{x^2}{2} \right]_0^2 + [x^2 - 2x]_2^4 = 2 + (8 - 0) = 10.$$

Visualizing the two parts of FTC together

- 14 For $g(x) = \int_0^x (t^2 - 2) \, dt$, find $g'(x)$ using FTC Part 1, then find where g has a local minimum by analyzing the sign of $g'(x)$.

$g'(x) = x^2 - 2$ (FTC Part 1). Setting $g'(x) = 0$: $x = \pm \sqrt{2}$. Since g' changes from negative to positive at $x = \sqrt{2}$ (for $x > 0$), g has a local minimum at $x = \sqrt{2}$.

A capstone FTC application

- 15 The rate of water flowing into a tank is $r(t) = 5 + 2\sin(t)$ liters/minute. Find the total volume that flows in during $[0, \pi]$ minutes, using FTC.

$$\text{Volume} = \int_0^{\pi} (5 + 2\sin t) \, dt = [5t - 2\cos t]_0^{\pi} = (5\pi + 2) - (0 - 2) = 5\pi + 4 \approx 19.71 \text{ liters}.$$

FTC Part 1 with more complex upper limits

- 16 Find $\frac{d}{dx} \int_0^{x^3} \sqrt{1+t^2} \, dt$ using the chain rule with FTC.

$$= \sqrt{1+(x^3)^2} \cdot 3x^2 = 3x^2 \sqrt{1+x^6}.$$

- 17 Find $\frac{d}{dx} \int_{x^2}^5 \cos(t) \, dt$, using $\int_{x^2}^5 = -\int_5^{x^2}$ before differentiating.

$$\int_{x^2}^5 \cos t \, dt = -\int_5^{x^2} \cos t \, dt, \text{ so the derivative is } -\cos(x^2) \cdot 2x = -2x\cos(x^2).$$

Net change and the FTC

- 18** A tank starts with 200 liters of water. Water flows out at rate $r(t) = 3t + 2$ liters/minute for the first 5 minutes. Use FTC to find the amount of water in the tank at $t = 5$.

Net change $= - \int_0^5 (3t + 2) dt = - \left[\frac{3t^2}{2} + 2t \right]_0^5 = - (37.5 + 10) = - 47.5$ liters. Amount remaining $= 200 - 47.5 = 152.5$ liters.

Visualizing FTC Part 2 as area

- 19** The graph shows $f(x) = x^2 - 1$ on $[0, 3]$. Use FTC to find $\int_0^3 (x^2 - 1) dx$ and explain why the result is not simply the area of the shaded region (part of the curve dips below the axis).

$\left[\frac{x^3}{3} - x \right]_0^3 = (9 - 3) - 0 = 6$. This is the SIGNED area: the small region below the axis on $[0, 1]$ is subtracted from the larger region above the axis on $[1, 3]$, so 6 is a net value, not the total unsigned area.

Combining FTC Part 1 and Part 2 in one problem

- 20** Let $F(x) = \int_1^x (3t^2 - 2) dt$. Find $F(2)$ using Part 2, then verify $F'(2)$ using Part 1 and confirm it matches direct substitution into the integrand.

$F(2) = [t^3 - 2t]_1^2 = (8 - 4) - (1 - 2) = 4 - (-1) = 5$. By Part 1, $F'(x) = 3x^2 - 2$, so

$F'(2) = 12 - 2 = 10$, which matches the integrand $3t^2 - 2$ evaluated at $t = 2$: $3(4) - 2 = 10$ ✓.